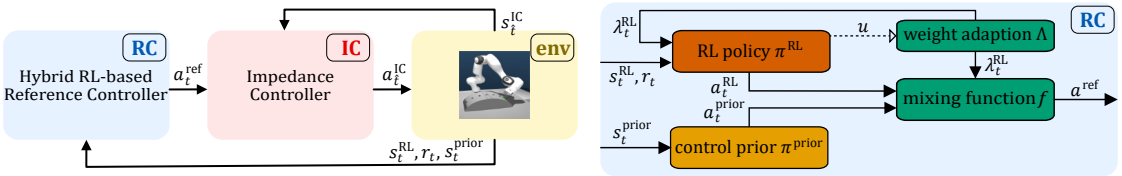


(a) (b)

Fig. 2. Panel (a) shows the complete control pipeline with higher-level reference controller RC and lower-level impedance controller IC. Panel (b) gives a detailed view of the RC, an adaptive hybrid RL agent based on CHEQ.

Nominal Control. The nominal controller outputs a_t^{prior}, including next desired position and orientation and fixed predefined impedance gains

$$a_t^{\text{prior}} = (\Delta p_t^{\text{prior}}, K^{\text{prior}}, \zeta^{\text{prior}}).$$

Following a simple and established approach, we define our reference path r^{path} by interpolating between 7 predefined via-points for the position (Lynch and Park, 2017) and using the SLERP method (Shoemake, 1985) for the orientation. Since our hybrid controller might be far from the defined path, we cannot define the motion control with a fixed velocity. Instead, we choose the next control point $p_t^{\text{prior,ref}}$ as the farthest point within a predefined radius.

CHEQ. CHEQ adaptively blends the action of the RL agent a_t^{RL} and the nominal controller a_t^{prior} (cf. Fig. 2b). The combined action a_t^{ref} is then a linear weighted sum

$$a_t^{\text{ref}} = (1 - \lambda_t^{\text{RL}}) \cdot a_t^{\text{prior}} + \lambda_t^{\text{RL}} \cdot a_t^{\text{RL}},$$

with the adaptive weight $\lambda_t^{\text{RL}} \in [0, 1]$. The weight is adapted based on the uncertainty $u(s_t^{\text{RL}}, a_t^{\text{RL}}, \lambda_t^{\text{RL}})$ of an ensemble of E Q-networks Q_{θ_e}, with $e = 1 \ldots E$. To simplify notation we introduce $z_t = (s_t^{\text{RL}}, a_t^{\text{RL}}, \lambda_t^{\text{RL}})$. The uncertainty is estimated as the standard deviation of the E critic predictions $u(z_t) = (\frac{1}{E}\sum_{e=1}^{E}(Q_{\theta_e}(z_t) - \mu(z_t))^2)^{-1}$ with $\mu(z_t) = \frac{1}{E}\sum_{e=1}^{E} Q_{\theta_e}(z_t)$. The weighting factor λ_t^{RL} is then a clipped linear function between a minimum uncertainty $u_{\min}$ and maximum uncertainty $u_{\max}$.

5. SIMULATION STUDY

We first conducted a simulation study to show that (i) variable gains benefit the polishing task and that (ii) CHEQ can learn such gains with safe and efficient exploration.

5.1 Evaluating the Need for VIC

To show the benefits of variable impedance, we evaluate the nominal controller with fixed and partly fixed gains.

Experimental Setup. We define a fixed gain controller with suitable but untuned gains as a baseline, calling this controller *Nominal (untuned)*. Using Bayesian Optimization (BO) in Weights&Biases (Biewald, 2020), we tuned this controller over 500 episodes and refer to it as *Nominal (tuned)*. Next, we partitioned the workpiece into 5 sections (see Fig. 1) and applied BO to optimize individual gains per section. This involved 500 episodes per section, iteratively using the best-found gains from previous sections. We call this *Nominal (tuned section-wise)*.

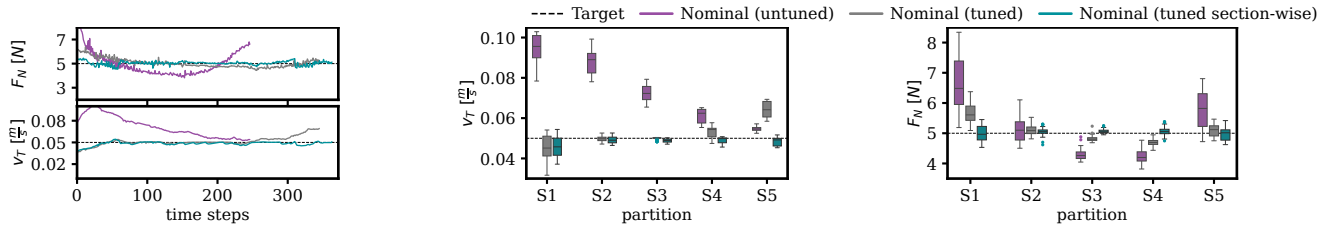
Results. Fig. 3 compares the force-velocity behavior of the three controllers over an episode, along with boxplots for each of the five object sections. The dotted line marks the force and velocity targets of 5 N and 0.05 m s^{-1}. The untuned controller fails to meet the targets, while the BO-tuned fixed gain controller performs significantly better but shows larger errors at the task's beginning and end, where the object curves steeply. The section-wise tuned controller effectively reduces errors in these regions, showing that variable impedance gains can yield superior performance. However, dividing the controller into arbitrary sections is impractical with manual methods. RL offers the potential to tune state-dependent gains.

5.2 Learning VIC with RL

Next, we evaluate the effectiveness of RL in learning state-dependent VIC. Our findings show that CHEQ not only enables safe exploration during training but also successfully learns effective polishing. In contrast, standard RL only learns when unsafe behavior is permitted.

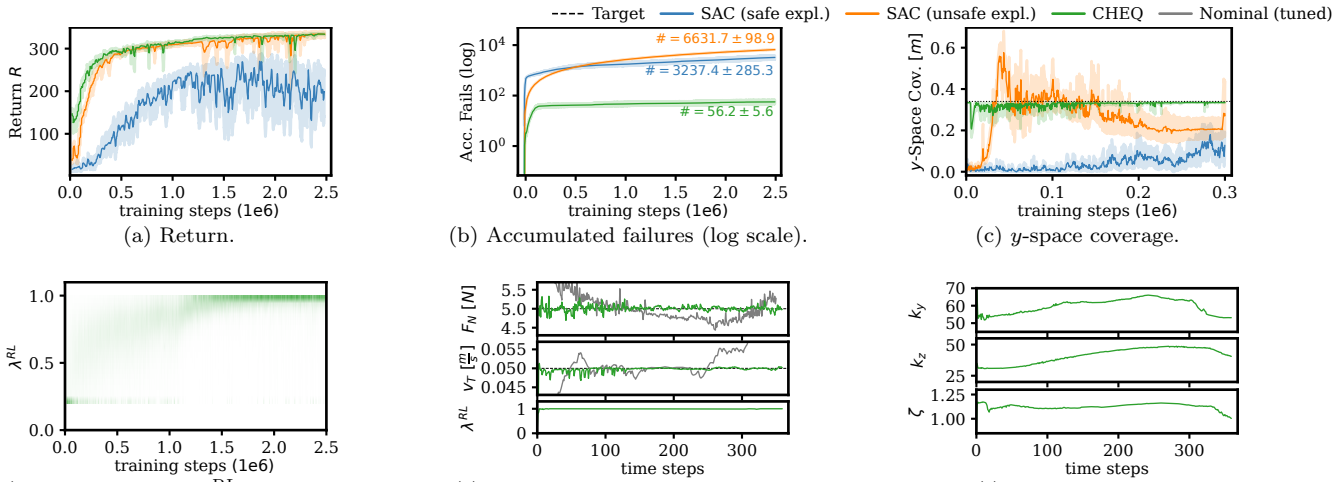
Experimental Setup. We compared two SAC variants, *SAC (unsafe exploration)* and *SAC (safe exploration)*, along with our hybrid *CHEQ* method. The latter two enforce safe exploration by applying velocity and force limits, bounding the end-effector's position, and constraining orientation to task-appropriate values (Details in Cramer et al. (2024b)). Episodes are truncated upon constraint violations to ensure minimal hardware wear and safeguard nearby humans. For *SAC (unsafe exploration)*, all restrictions are lifted. The CHEQ agent uses the *Nominal (untuned)* controller as its prior, arguing that an easy-to-define, untuned controller suffices for hybrid RL.

Results. Figures 4a-c show the performance of RL agents during training, averaged over 10 runs with 95 % quantiles. *SAC (safe exploration)* achieves low returns, failing to learn the task effectively. In contrast, *SAC (unsafe exploration)* learns the task but accumulates over 6×10^3 failures. We count only the first violation per episode, though actual failures are much higher, as we do not truncate episodes. This high failure rate makes it unsuitable for hardware due to excessive wear and tear. The *CHEQ* agent achieves high returns, slightly outperforming unsafe SAC while reducing failures by three orders of magnitude. Differences between approaches are evident in the y-space coverage (Fig. 4c). Since the polishing path is along the y-axis (see Fig. 1), y-space coverage is a proxy for task space exploration. In the first 300×10^3 training steps, the safe SAC agent covers less than half the task, as safety constraints and random exploration limit progress. The unsafe SAC agent covers the full space but overshoots the endpoint. For CHEQ, the nominal controller guides exploration efficiently towards the full task space, and the



(a) Force-velocity behavior. (b) Velocity behavior per section. (c) Force behavior per section.

Fig. 3. Comparison of force and velocity over one episode (a) and boxplots for the five sections (b,c). The fixed-gain controller tuned with BO shows increased performance, and tuning per section further improves behavior.



(a) Return. (b) Accumulated failures (log scale). (c) y-space coverage.

(d) Weighting factor λ^{RL} distribution. (e) Force-velocity-lambda behavior. (f) Impedance gains.

Fig. 4. Comparison of return, fails and y-space coverage of CHEQ and SAC variants(a-c). CHEQ achieves highest performance with only 56 fails. Over training, the distribution of the weighting factor λ^{RL} rises to 1 (d). Fig. e and f show the trained agent after 2.5×10^6 steps. The agent achieves good polishing behavior by adapting the gains.

small λ^{RL} at the beginning ensures the agent uses just enough random exploration to learn the task safely and efficiently. Looking at the λ^{RL}-distribution (see Fig. 4d), the agent first learns a small residual on top of the nominal controller. This already provides good polishing behavior and high returns. Over time, the agent transitions to full RL control, continuously improving performance. This induces a learning curriculum, ensuring safe exploration. Compared to the tuned baseline (Fig. 4e), CHEQ achieves superior force and velocity tracking by adapting gains dynamically throughout the task (Fig. 4f). A complete comparison with all priors is in Cramer et al. (2024b). Overall, CHEQ demonstrates significantly safer learning than standard RL while outperforming the tuned baseline.

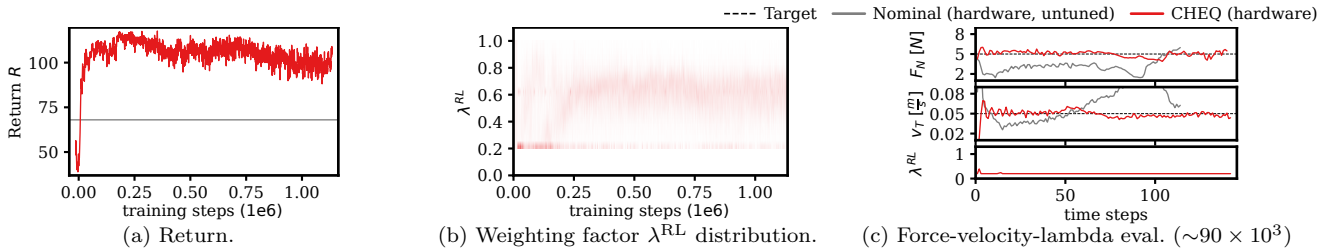
6. CASE STUDY: REAL-WORLD 3D POLISHING

In this section, we address the challenges and solutions of training CHEQ on hardware, demonstrated through a 3D polishing task with a robotic manipulator. This task, representative of contact-rich manufacturing problems, marks CHEQ's first successful hardware implementation.

Experimental Setup. Additional challenges need to be considered to achieve stable training on hardware. On our hardware setup, we can achieve a maximum control frequency of 20 Hz while simultaneously dealing with measurement noise in the robot state and force measurements. Further, our simulation results revealed one critical aspect specific to CHEQ. In the initial training phase, the weighting factor fluctuates greatly between time steps, resulting in highly fluctuating IC gains. In simulation, this does not pose a problem. On real hardware, however, the gain fluctuations lead to a chattering end-effector motion. This doubly affects training as it complicates the control and increases the motion-dependent force sensor noise. To mitigate this, we use an average of 10 time steps to compute our uncertainty, leading to smoother weights. Combining the mentioned challenges, we find that data collected on our hardware is less reliable than in simulation. This might lead to unstable training. CHEQ provides a mechanism to mitigate this by slowing the curriculum. Thus, for our hardware experiments, we set more conservative uncertainty limits. This reduces the agency of the RL agent in the beginning and leads to more stable, safer, albeit slower, learning. We use an untuned prior (*Nominal hardware (untuned)*) and set a UTD ratio of 2 to account for the higher cost of environment steps on hardware.

Results. On hardware, the return improves quickly, surpassing the *Nominal hardware (untuned)* controller after 6.5×10^3 steps. Even this early in the training process (250×10^3 steps and 8 hours of training), good polishing behavior is achieved. The trained RC maintains smooth force and velocity close to the targets, achieving this with a low weight around 0.2 (see Fig. 5c). Importantly, this behavior is achieved while accumulating only five failures.

Longer training does not further improve the return (cf. Fig. 5a). In our adaptive setting, the agent must learn the complete state, action, and weight space. Whenever the agent learns a higher weight, the return drops and rises



(a) Return. (b) Weighting factor λ^{RL} distribution. (c) Force-velocity-lambda eval. ($\sim 90 \times 10^3$)

Fig. 5. Return (a) and λ^{RL}-distribution (b) of CHEQ on hardware. The agent achieves good polishing behavior (c).

again once the agent has mastered this part. With the slow curriculum, the agent has not yet learned the complete weight space. Still, we see no benefit in further training, as our early results show that a small residual on the untuned controller suffices for effective polishing. Even though a residual term is enough to learn good polishing behavior in this setting, CHEQ still works differently than traditional fixed-weight approaches. Cramer et al. (2024a) demonstrated that a fixed weight compromises safe learning and performance; comparable final performance can only be achieved with the optimal weight for a certain scenario and control prior. However, even then, training benefits from the curriculum induced by adaptive weighting. We conclude that CHEQ learns good polishing behavior safely and efficiently on hardware.

7. CONCLUSION

This work applied the CHEQ algorithm, a novel AHRL algorithm, to robotic polishing [1] . In a simulation study, we highlighted the benefits of variable impedance and showed that CHEQ outperforms standalone RL. The adaption mechanism ensured safe exploration by gradually transitioning from control priors to autonomous operation. On hardware, CHEQ achieved effective polishing performance. This study demonstrates the potential of AHRL for hardware training, combining safety with performance.

ACKNOWLEDGEMENTS

We thank Johannes Berger, Aditya Pradhan, Lukas Wildberger, and Furkan Celik for their help with the hardware implementations and input during the early stages of this research. We also thank Alexander Gräfe for the many helpful discussions. This work was funded in part by the German Federal Ministry of Education and Research ("Demonstrations- und Transfernetzwerk KI in der Produktion (ProKI-Netz)" initiative, grant number 02P22A010). Computations were performed with computing resources granted by RWTH Aachen University under the projects *thes*1735, *p*0022348, *p*0022301, and *p*0021919.

REFERENCES

Anand, A.S., Hagen Myrestrand, M., and Gravdahl, J.T. (2022). Evaluation of Variable Impedance- and Hybrid Force/MotionControllers for Learning Force Tracking Skills. In *IEEE Int. Symp. on Syst. Integr.*, 83–89.

Anand, A.S., Kaushik, R., Gravdahl, J.T., and Abu-Dakka, F.J. (2024). Data-Efficient Reinforcement Learning for Variable Impedance Control. *IEEE Access*, 12, 15631–15641.

Biewald, L. (2020). URL `https://www.wandb.com/`.

Bogdanovic, M., Khadiv, M., and Righetti, L. (2020). Learning variable impedance control for contact sensitive tasks. *IEEE Rob. and Aut. Letters*, 5(4), 6129–6136.

Buchli, J., Stulp, F., Theodorou, E., and Schaal, S. (2011). Learning variable impedance control. *Int. J. of Rob. Res.*, 30(7), 820–833.

Büchler, D., Guist, S., Calandra, R., Berenz, V., Schölkopf, B., and Peters, J. (2022). Learning to Play Table Tennis From Scratch Using Muscular Robots. *IEEE Trans. on Rob.*, 38(6), 3850–3860.

Ceola, F., Rosasco, L., and Natale, L. (2024). RESPRECT: Speeding-up Multi-Fingered Grasping With Residual Reinforcement Learning. *IEEE Rob. and Aut. Letters*, 9(4), 3045–3052.

Cheng, R., Verma, A., Orosz, G., Chaudhuri, S., Yue, Y., and Burdick, J. (2019). Control Regularization for Reduced Variance Reinforcement Learning. In *Int. Conf. on Mach. Learn.* PMLR.

Cramer, E., Frauenknecht, B., Sabirov, R., and Trimpe, S. (2024a). Contextualized Hybrid Ensemble Q-learning: Learning Fast with Control Priors. In *Reinforcement Learning Journal*, volume 2, 926–945. Amherst Massachusetts.

Cramer, E., Jäschke, L., and Trimpe, S. (2024b). CHEQ-ing the Box: Safe and Efficient Variable Impedance Learning for Robotic Polishing (extended version). URL `http://tiny.cc/cramer_polishing_2024`.

Davchev, T., Luck, K.S., Burke, M., Meier, F., Schaal, S., and Ramamoorthy, S. (2022). Residual Learning From Demonstration: Adapting DMPs for Contact-Rich Manipulation. *IEEE Rob. and Aut. Let.*, 7(2), 4488–4495.

Fiedler, K.H. (1998). Processing (Grinding and Polishing). In *The Properties of Optical Glass*, 245–262. Springer.

Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Int. Conf. on Mach. Learn.* PMLR.

Johannink, T., Bahl, S., Nair, A., Luo, J., Kumar, A., Loskyll, M., Ojea, J.A., Solowjow, E., and Levine, S. (2019). Residual reinforcement learning for robot control. In *Int. Conf. on Rob. and Aut.(ICRA)*, 6023–6029. IEEE.

Khatib, O. (1995). Inertial Properties in Robotic Manipulation: An Object-Level Framework. *Int. J. of Rob. Res.*, 14(1), 19–36.

Luo, J., Hu, Z., Xu, C., Tan, Y.L., Berg, J., Sharma, A., Schaal, S., Finn, C., Gupta, A., and Levine, S. (2024). SERL: A Software Suite for Sample-Efficient Robotic Reinforcement Learning.

Luo, J., Solowjow, E., Wen, C., Ojea, J.A., Agogino, A.M., Tamar, A., and Abbeel, P. (2019). Reinforcement Learning on Variable Impedance Controller for High-Precision Robotic Assembly. In *Int. Conf. on Rob. and Aut.(ICRA)*, 3080–3087.

Lynch, K.M. and Park, F.C. (2017). *Modern Robotics: Mechanics, Planning, and Control.* Cambridge University Press.

Martín-Martín, R., Lee, M.A., Gardner, R., Savarese, S., Bohg, J., and Garg, A. (2019). Variable impedance control in end-effector space: An action space for reinforcement learning in contact-rich tasks. In *IEEE Int. Conf. on Int. Rob. and Sys. (IROS).*

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., and Ostrovski, G. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529.

Rana, K., Dasagi, V., Haviland, J., Talbot, B., Milford, M., and Sünderhauf, N. (2023). Bayesian controller fusion: Leveraging control priors in deep reinforcement learning for robotics. *Int. J. of Rob. Res.*, 42(3), 123–146.

[1] `github.com/Data-Science-in-Mechanical-Engineering/polishing-cheq`.